

A purely metric parallel-sum formula for infimal convolution on geodesic products

Candidate full answer for arXiv:2503.12939

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Status: candidate full answer, likely valid. Weighted product metrics on arbitrary finite products of geodesic spaces have an explicit metric infimal convolution, which is again a finite nondegenerate geodesic distance. This gives a broad affirmative answer to the source’s existential question, subject to human review.

1 Source question

The source is Nicolò De Ponti, Giacomo Enrico Sodini, and Luca Tamanini, *The infimal convolution structure of the Hellinger-Kantorovich distance*, arXiv:2503.12939v1 [1]. After showing that metric infimal convolution can be degenerate or fail the triangle inequality, and after discussing the more structured normed-space case, the authors ask in the Introduction (page 5):

“This raises the compelling question of whether there exist other purely metric examples, similar to the Hellinger–Wasserstein case, where this construction yields meaningful distances.”

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show that this construction leads to a symmetric, non-negative function, though it can be degenerate (i.e., being null between different points) and may fail to satisfy the triangle inequality. A more structured behavior emerges in the case the inf-convolution is performed between (Hilbertian) norms, where the problem aligns naturally with the infimal convolution in convex analysis [38]. This raises the compelling question of whether there exist other purely metric examples, similar to the Hellinger–Wasserstein case, where this construction yields meaningful distances.

In another direction, we also thoroughly examined the properties of the single-step minimization problem (1.12) between the Wasserstein and Hellinger distances. This problem corresponds to an iteration step of the associated Minimizing Movement scheme, making it interesting to explore how our analysis could be connected to the theory of gradient flows [9]. Notice that a peculiarity of our situation is that

Figure 1: The source question in its full page-width context, page 5.

For two costs c_1, c_2 on a set Z , the source defines the square of their metric infimal convolution at z^0, z^1 by

$$(c_1 \nabla c_2)^2(z^0, z^1) = \liminf_{N \rightarrow \infty} \inf_P N \sum_{i=1}^N (c_1^2(x_{i-1}, y_i) + c_2^2(y_i, x_i)), \quad (1)$$

where $P = (x_0, \dots, x_N; y_1, \dots, y_N)$, $x_0 = z^0$, and $x_N = z^1$.

2 Proof intuition

On a product of geodesic spaces, give each factor one weight in the first metric and another weight in the second. During an alternating step, each coordinate can place its intermediate point at its own optimal fraction along a geodesic. Minimizing the two weighted squared lengths produces the familiar parallel-sum coefficient

$$(a_1^{-2} + a_2^{-2})^{-1/2}.$$

The same elementary quadratic inequality gives a lower bound for every possible path. Thus the effective product distance is not merely the limit: its square is the exact minimum energy at every discretization level N .

3 The product theorem

Theorem 1 (Purely metric parallel sum). *Let (X_k, d_k) , $k = 1, \dots, m$, be geodesic metric spaces and let $Z = \prod_{k=1}^m X_k$. For positive numbers a_{jk} , $j = 1, 2$, define metrics*

$$\rho_j(z, z')^2 = \sum_{k=1}^m a_{jk}^2 d_k(z_k, z'_k)^2, \quad j = 1, 2. \quad (2)$$

Set

$$a_{*k} := (a_{1k}^{-2} + a_{2k}^{-2})^{-1/2} = \frac{a_{1k} a_{2k}}{\sqrt{a_{1k}^2 + a_{2k}^2}} \quad (3)$$

and

$$\rho_*(z, z')^2 := \sum_{k=1}^m a_{*k}^2 d_k(z_k, z'_k)^2. \quad (4)$$

Then

$$\rho_1 \nabla \rho_2 = \rho_*.$$

More precisely, for every $N \geq 1$ and every pair of endpoints, the infimum of the N -path energy in (1) is exactly the square of their ρ_* -distance.

Proof. Each function in (2) and (4) is a metric. This follows from the coordinate triangle inequalities and Minkowski's inequality. They are geodesic: choose constant-speed geodesics in every factor and run them with a common time parameter.

Fix endpoints z^0, z^1 and an arbitrary N -path $P = (x_0, \dots, x_N; y_1, \dots, y_N)$ between them. For each i, k , put

$$A_{ik} = d_k(x_{i-1,k}, y_{i,k}), \quad B_{ik} = d_k(y_{i,k}, x_{i,k}), \quad C_{ik} = d_k(x_{i-1,k}, x_{i,k}).$$

The triangle inequality gives $C_{ik} \leq A_{ik} + B_{ik}$. For positive a, b and nonnegative A, B ,

$$a^2 A^2 + b^2 B^2 - \frac{a^2 b^2}{a^2 + b^2} (A + B)^2 = \frac{(a^2 A - b^2 B)^2}{a^2 + b^2} \geq 0. \quad (5)$$

Applying (5) coordinatewise yields

$$\begin{aligned} \rho_1^2(x_{i-1}, y_i) + \rho_2^2(y_i, x_i) &= \sum_k (a_{1k}^2 A_{ik}^2 + a_{2k}^2 B_{ik}^2) \\ &\geq \sum_k a_{*k}^2 (A_{ik} + B_{ik})^2 \\ &\geq \sum_k a_{*k}^2 C_{ik}^2 = \rho_*^2(x_{i-1}, x_i). \end{aligned}$$

Consequently, Cauchy–Schwarz and the triangle inequality for ρ_* give

$$\begin{aligned}\mathcal{E}_N(P) &\geq N \sum_{i=1}^N \rho_*^2(x_{i-1}, x_i) \\ &\geq \left(\sum_{i=1}^N \rho_*(x_{i-1}, x_i) \right)^2 \geq \rho_*^2(z^0, z^1).\end{aligned}\tag{6}$$

It remains to attain the lower bound. For each coordinate choose a constant-speed geodesic $\gamma_k : [0, 1] \rightarrow X_k$ from z_k^0 to z_k^1 , and write $D_k = d_k(z_k^0, z_k^1)$. Define

$$x_{i,k} = \gamma_k(i/N), \quad t_k = \frac{a_{2k}^2}{a_{1k}^2 + a_{2k}^2}, \quad y_{i,k} = \gamma_k\left(\frac{i-1+t_k}{N}\right).\tag{7}$$

Then

$$A_{ik} = \frac{t_k D_k}{N}, \quad B_{ik} = \frac{(1-t_k) D_k}{N}.$$

The choice of t_k is the equality case in (5), so every alternating step in (7) has cost

$$\sum_k (a_{1k}^2 A_{ik}^2 + a_{2k}^2 B_{ik}^2) = \frac{1}{N^2} \sum_k a_{*k}^2 D_k^2 = \frac{\rho_*^2(z^0, z^1)}{N^2}.$$

Multiplying the sum of the N step costs by N gives $\mathcal{E}_N(P) = \rho_*^2(z^0, z^1)$. Together with (6), this proves the exact minimum formula for every N , and hence the asserted infimal convolution identity. $\square$

4 A branching, non-vector example

Let T be the metric tripod obtained by gluing three copies of $[0, 1]$ at their zero endpoints, with path distance d_T . On $Z = T \times [0, 1]$, define

$$\begin{aligned}\rho_1((x, s), (y, t))^2 &= d_T(x, y)^2 + 4|s - t|^2, \\ \rho_2((x, s), (y, t))^2 &= 4d_T(x, y)^2 + |s - t|^2.\end{aligned}$$

In each coordinate the effective weight is

$$(1^{-2} + 2^{-2})^{-1/2} = \frac{2}{\sqrt{5}}.$$

Theorem 1 therefore gives

$$(\rho_1 \nabla \rho_2)((x, s), (y, t)) = \frac{2}{\sqrt{5}} \sqrt{d_T(x, y)^2 + |s - t|^2}.\tag{8}$$

This is a finite nondegenerate geodesic distance on a space with genuine branching. The inputs are not proportional: on tripod fibers $\rho_2 = 2\rho_1$, whereas on interval fibers $\rho_1 = 2\rho_2$. Thus both inputs contribute nontrivially to the output.

5 Verification, scope, and novelty

The included exact checker expands (5), verifies the optimizing fraction and effective coefficient symbolically, checks the tripod constants, and tests the lower bound on deterministic rational product paths. Run

```
conda run --no-capture-output -n sandbox python \
  code/check_parallel_sum.py
```

from the packet directory. Computation is only an audit; the proof above is exact.

The result answers the source’s existential question affirmatively with a large nonlinear class. It does not classify every well-behaved pair, prove that arbitrary geodesic metrics have a distance-valued infimal convolution, or address the separate question about p -Hellinger–Kantorovich distances.

Before promotion, the run indexes were searched using arXiv:2503.12939, the source title, the exact question terms, metric infimal convolution, geodesic products, weighted product metrics, and parallel sums. A bounded arXiv/web search used the exact source question and close product/geodesic/parallel-sum phrases. It found the source paper and unrelated functional infimal-convolution literature, but no separate statement of Theorem 1. The current arXiv record is v1, submitted 17 March 2025, and still contains the question. This records the search bounds and is not an exhaustive novelty certification.

Human review is recommended. The main scope question is whether this broad existential answer matches what the source authors intend by “purely metric”; mathematically, the highest-value checks are the scalar equality case and the coordinatewise attaining path (7).

References

- [1] N. De Ponti, G. E. Sodini, and L. Tamanini, *The infimal convolution structure of the Hellinger-Kantorovich distance*, arXiv:2503.12939v1, 2025. <https://arxiv.org/abs/2503.12939>.